



Using Video Overlay for Communication of Mine Site Opportunities

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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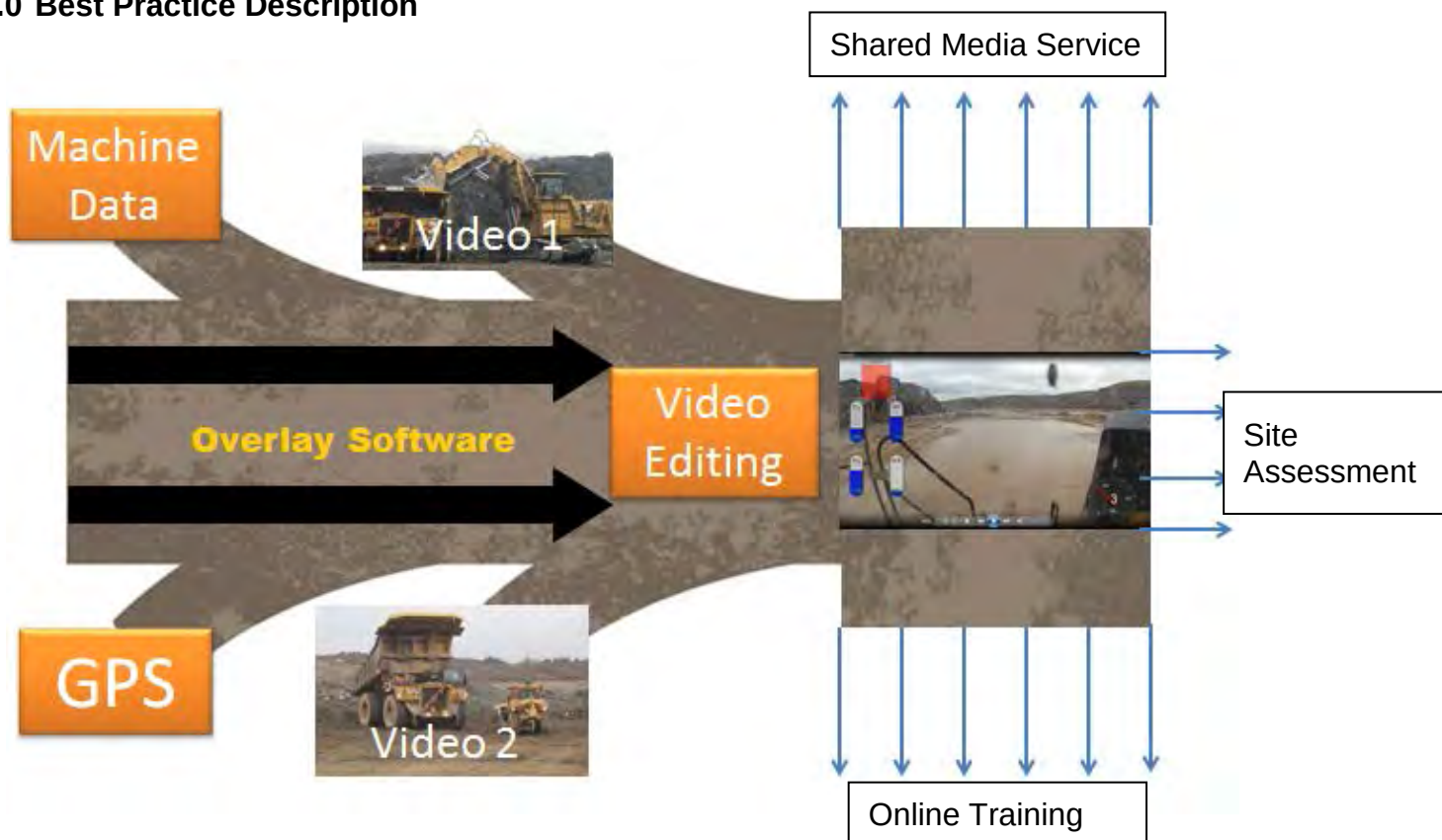
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1.0 Introduction

Haul road design and maintenance is an important part of a mine site's overall performance. Not only does it impact cycle times and productivity, but also has a bearing on the life of truck components such as the transmission, duo-cone seals and most costly, the machine frame. It is often difficult to explain the long-term effects of haul road conditions with pictures, FELA numbers or R.A.C. data alone. This Best Practice outlines a method to produce videos overlaid with gauges and graphs showing live data as experienced directly by a truck travelling on the haul road or dump area. It should be used in conjunction with other haul road improvement analysis tools by mine site reliability engineers, dealer MARC managers and application engineers to provide mine site managers with a visual "as is" picture of what a machine is experiencing on a day to day basis on *their* own haul roads.

Videos produced with this Best Practice can also be produced and included in the Application section of a mine site assessment when reviewing haul roads with mine management. Videos can then easily be shared with the enterprise to sensitize people to specific haul road issues such as slope consistency, speed on grade, overloading, drainage or even truck-shovel placement.

2.0 Best Practice Description



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The key Elements of an effective video involve:

- Defining the Purpose of the Video for the selected minesite
- Acquiring the right hardware for onsite filming and data collection
- Planning the video filming and data acquisition with the minesite
- Setting up on the machine and collecting the data
- Using overlay and editing software to produce a quality video
- Distributing the message

3.0 Implementation Steps

3.1 Defining the Purpose of the Video for the selected minesite

This is the most important step in the process of delivering an effective message to the audience. Regardless of the quality of the data and footage achieved, the purpose of the video must be defined and understood. Agreeing on a set purpose the mine management will also help securing access to machines and other resources during filming. Haulroad maintenance may not be the only priority issue and this tool can help explain many other situations

The following are some questions to begin with:

1. Is there a clear burning platform?
 - We need to communicate to operators to slow down during the fall in order to reduce tire cuts.
 - There is a lot traffic on the haul road, other vehicles are not yielding to production trucks, causing longer cycle times.
 - We are reporting many RAC events and we believe the haul road in the 'C' pit is poorly maintained.
2. Are there known situations that have been reported?
 - Overspills, large boulders left in the road
 - Water, ineffective drainage, puddles
 - Operator discomfort due to dips and harsh turns
 - Truck Bunching, long wait times at the loading areas
 - Unequal loads, bailing of material
 - Contact with the berm
 - Tire cuts, tire wear or delamination
 - Incidents, collisions, or close calls in some areas of the haul road
3. Are there two conditions to compare?
 - Good haul road versus poorly maintained haul road
 - Good use of the retarder versus improper use of the retarder
 - Speed on grade of different operators, different truck models

Answering these questions will allow better planning when selecting the correct framing of the videos and the limited number of data channels to sample. Don't assume that all you will need is a shot from the front of the machine. A camera tucked under the truck can capture the stress a strut goes under running on a poorly maintained road. Another camera facing an operator's seat can capture how much bouncing around the operator goes through during a cycle.

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3.2 Acquiring the right hardware for onsite filming and data acquisition

3.2.1 Video Cameras



Figure 1 GoPro Hero 2

The most popular heavy duty camera platform at the time of writing this Best Practice is the GoPro series of cameras. All our videos to date have been filmed with several GoPro cameras. These are designed for extreme sports and the *motorsports* mounts are a good fit for mining and truck mounting. If GoPro cameras are not available, look for cameras with the following features:

- Compatibility with versatile mounting systems
- Availability of accessories (online, local store)
- Battery life
- Picture Quality

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3.2.2 Mounting hardware



Figure 2: Motorsports Accessory and Handlebar Accessory

Within the GoPro series of mounts is the *motorsports accessory* which uses a strong vacuum cup and works well when mounting on flat parts of the machine such as windows and smooth painted surfaces. The length of the arm included with the accessory allows the user to position the camera easily. You can also use the handlebar accessory shown on the left of Figure 2. This mount secures to piping such as Catwalks. Make sure to secure the mount to the lower parts of the catwalk to avoid excessive vibration. You will also need to use extension arms in order to pivot the camera correctly. The GoPro sliding mount can also be adapted to fit to common $\frac{1}{4}$ " threaded camera mounts as shown in Figure 3 where a $\frac{1}{4}$ " nut is stuck onto the back of a standard Gopro camera receptacle. This mount is useful when trying to reach out to an area clear of the mounting point, such as over the top of the canopy.



Figure 3: Standard $\frac{1}{4}$ " threaded Camera Accessories and adaptor

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3.2.3 Useful accessories

- **Extra Batteries**



While one battery will only last little more than one hour, the battery extensions will prolong the footage for up to 3 hours

- **Wi-Fi pack**



- The above batteries for the Hero 2 allow to connect the camera either to a remote dongle to trigger the camera and monitor its memory space and battery or to a smartphone which will additionally provide a preview of the shot. Note that the recent Hero 3 Black edition has this feature built in.



- **Fog inserts**

In humid conditions, the front lens will fog up and blur the image. Regardless of how dry you keep the equipment, any residual moisture will condense on the front lens. This can easily be avoided with fog inserts, these are white powdery strips that are inserted in the sides of the camera housing and absorb the humidity before it condenses on the front lens. They can be reused after drying in a warm oven for a few minutes. You can also make your own fog inserts by folding up a paper towel six

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times over, placing the folded strip in a warm oven to dry and placing in the camera housing



- **Camera bag**

The GoPro cameras have a tendency to turn on when they bounce around. This means when you're ready to take a video, the battery is often drained out. A small padded camera bag will prevent this from happening as well as protect the gear.



- **Memory cards**

Often overlooked, plan for at least 16GB cards in each camera, hi-resolution files will fill the cards up very quickly. At 1080P, the camera will record a succession of 2GB files with about 8 minutes of recording. Use a sharpie and some masking tape to note the time and machine identification information on each

3.2.4 Data acquisition hardware

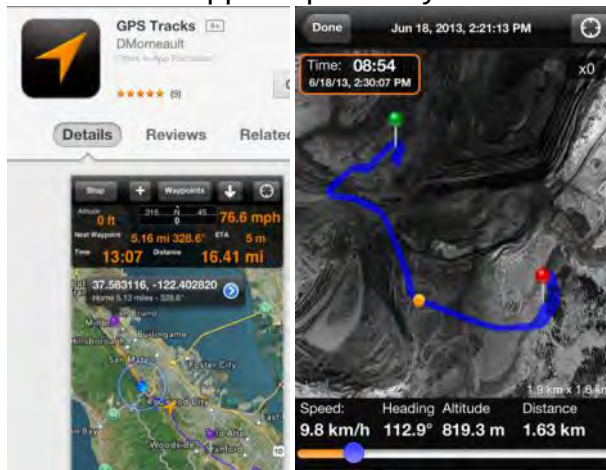
The hardware below will allow you to gather the data in raw form to be used with the overlay software. We have typically used Caterpillar Electronic Technician with a Comm adapter to link to the machine's ECM's. The software used to overlay the data is also capable of plotting the position of the machine as well as graphing acceleration forces captured by another device. There are inexpensive phone applications that are easy to use that can gather this data if needed. We have successfully used *GPS Tracks App* for iPhone by DMorneault to gather GPS position and combine in *RaceRender 2* to map out the course where the machine is located.

- Laptop with Electronic Technician and an active license, Power Inverter, Comm Adapter

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- GPS Tracks app for Iphone by DMorneault



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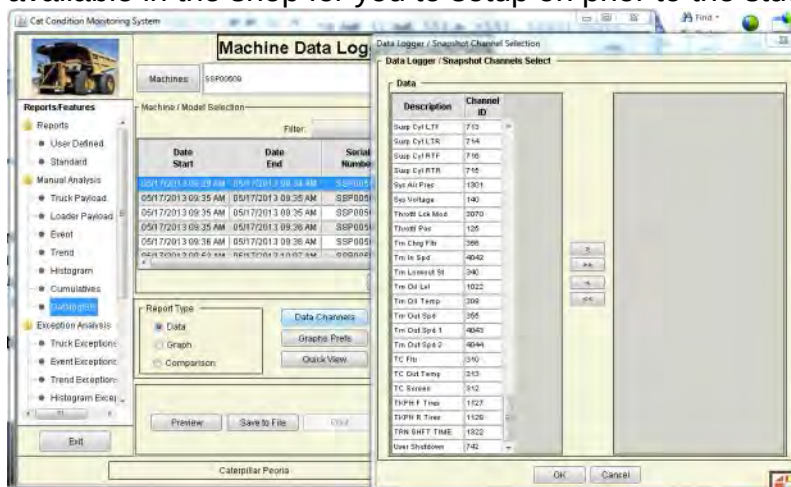
3.3 Planning each task with the minesite

Planning is also a very important part of this process, a minesite is a large complex operation and a careful plan will mitigate the effect of the study on production. A single truck out of normal production represents a large investment therefore setup time should be kept at a minimum. Below is an example of a typical schedule.

8:00 or Shift Change	Meet with foreman and operators that will be affected
8:30	Travel to machine meet point
9:45	Install camera gear, setup datalogger
10:00	Begin first take
11:00	End first take , Check data
11:15	Begin Second take
12:00	End Second take

When meeting with the foreman, take note of any exceptions that are made to regular operating practices such as maximum speed, maximum gear, target payload. The mine may choose to exceptionally raise or lower machine speeds or request the operator perform his task differently than usual. They may also remove all traffic from the selected route to measure the effect of a clear road or remove unwanted stoppages.

You will need about fifteen minutes to mount the camera gear on the machine as well as to setup the laptop, choose the datalog channels and setup the sampling rate and duration. If you are onsite for a few days for a site assessment, try selecting a day with no rain/snow as mud will cover the camera and snow may obstruct the view. Try to take into account the sun's position at that time to avoid glare that could make the picture unusable. It is important to know what ECM data channels you will be sampling ahead of time as it can take much time to go through the list. Machines today have over one hundred channels to choose from and you will need to select the right channels that will provide valuable information. For a list of available ECM channels, you can look them up for your selected machine in CMS at <https://catcms.cat.com> under Manual Analysis and Datalog/SS, or contact the DSN or technical communicator for a list. You can also check with the mine if a similar machine is available in the shop for you to setup on prior to the study day.



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3.4 Setting up on the machine and collecting the data

The day before filming, make sure that you charge all the batteries for the cameras and store them where they are unlikely of being inadvertently turned on. Check that Caterpillar Electronic Technician loads up well and that your license is active. Fully charge your laptop and your phone. Put the cameras, mounts, comm adaptor and laptop in a bag with shoulder straps.

Taking the video and datalog

This is the easiest part, but first use every precaution when working around a machine:

- Always use three points of contact
 - Carry all your gear in a shoulder bag, you will not have any free hands as you climb on the machine
 - Make sure the operator acknowledges your presence and is aware of what you are doing, request a radio if possible
 - Follow the machine lock-out processes for the areas of the machine you are addressing
4. Mount the camera according to the type of mounting hardware selected, do not start the camera. Consider weather mud will reach the camera. If the area surrounding the mounting location looks clean, chances are the camera will be clean for a while
 5. Preview the shot (optional). If you have connected a phone with the preview app, position the camera for the optimal framing. Figure 4 shows a typical setup
 6. Remove the camera from the housing. In order to start the datalogger and the camera at the same time, you should take the camera in the cab with you. Make sure you don't move the housing so as not to change the framing
 7. Setup Electronic Technician. Figure 5 shows a screenshot of a typical setup,
 - Set sample rate to 240 per minute
 - Set Duration to 60 minutes or more, this must be long enough to allow the time for a full cycle plus the time to get to the start of the cycle
 - Set the Pre-Trigger to 0, we will be starting the datalog in sync with the camera; a pre-trigger will insert a delay and must be removed.
 - Note the location of the datalogger file and name for later
 8. Select the data channels. Figure 6 shows channels for overlaying haul road information. You can save the list of channels in ET and it will be available when you select the same ECM again on that machine.
 9. Setup your GPS tracker (optional). If you're including a map and the position of the machine in the overlay design, setup the GPS logger according to the instructions in the user manual. A sample rate of 1 second should be enough, make sure the accuracy is acceptable, within ± 10 meters is enough for a haul road application. Figure 7 shows the start screen of the app.

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10. **Critical:** Start all three processes at the same time, press start on the phone, start on the ET screen, and press down on the top camera button all at once. You can sync up the timing of the video and the data (GPS and ET) later within RaceRender, however starting all three at once will save much time. Have the machine operator help you and countdown. If this is not possible for you, take careful note of the time difference between the start of the video and the start of the datalog.
11. Place your phone and laptop in a secure location, with the phone facing the window and preferably clipped to the visor or cab liner. Be careful not to close the laptop lid or pull the USB cable out.
12. Place the camera back into the housing
13. Start moving the machine. At this point you can let the operator go alone, or accompany him/her to make sure the devices don't fall or stop. Keep track of time, as the datalogger will stop recording once its time has run out.
14. At the end of the cycle, you can stop the datalogger, GPS recording and camera. You do not have to stop every device at the same time.
15. Check that the files have saved successfully. Check that the files open properly, you may need to do another take,
16. Travel to your drop point and collect all your gear



Figure 4

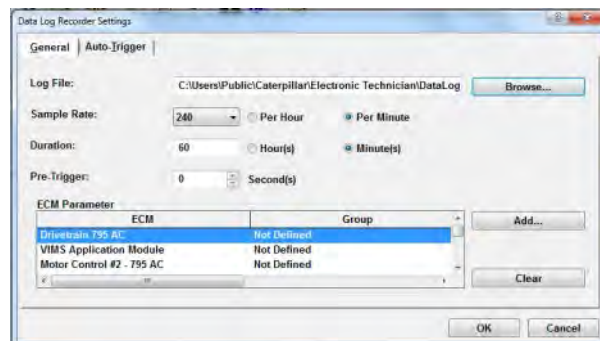


Figure 5

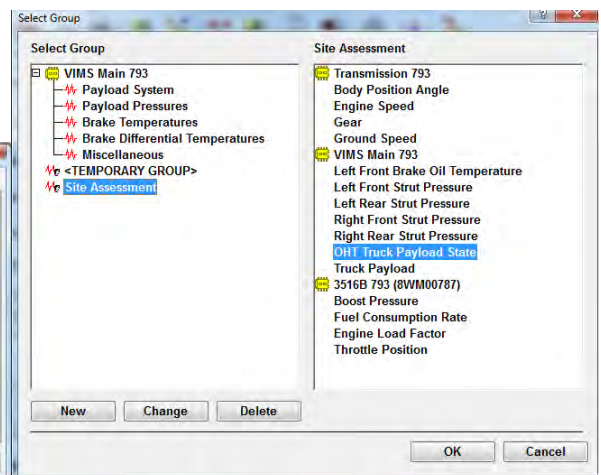


Figure 6

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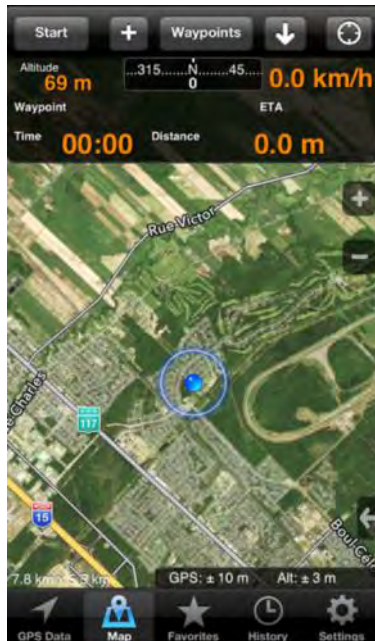


Figure 7

3.5 Using overlay and editing software to produce a quality video

Refer to the PowerPoint Document Titled: *Using RaceRender to Overlay machine data*

3.6 Distributing the message within the proper medium

When carefully planned, the filming and data acquisition should not take longer than half a day, leaving enough time to create at least a draft of the video within a full day for presenting at an approval meeting the next day. The mine management may want to make some changes to the displayed information to better suit their audience. Take this feedback into consideration when preparing the final version of the video.

Caterpillar uses <https://videoatwork.cat.com>, a media sharing website to securely share media such as these videos with internal Caterpillar and dealer personnel. The file will likely be larger than 500MB and can be sent with Caterpillar's Secure Internet File Transfer tool at <https://SIFT.cat.com> for internal Caterpillar and dealer transfers. For sharing the video with customers, leave a copy of the video with the customer in order for them to share within the organization. Respect the customer's data privacy protocols.

4.0 Resources Required

- Time:
 - o Half a day prior the shooting to plan, meet with the mine site, charge batteries and gather equipment
 - o Depending on the length of time for a cycle, plan on at least two takes, plus 20 minutes to mount the hardware, log into Caterpillar Electronic Technician and setup the datalogger. Usually no more than 3 hours total per segment.
 - o Setting up of the overlay design can take up to 8 hours, but a draft can be provided within 2 hours

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- o Depending on the speed of the computer used, the overlay process is done frame by frame once launched and can take from 20 minutes to several hours on an older laptop
- Personnel
 - o Machine Operator
 - o Camera/Datalogger operator - This person should ideally remain in the machine to ensure connections are working and datalogger is collecting data without interruption
- Cameras and accessories – 2 or more
- Camera mounting hardware
- GPS position tracking tool (either VIMS GPS, separate GPS, or phone app such as GPS Tracks for iPhone)
- Lock and key for machine lockout

5.0 Supporting Attachments / References

- Using RaceRender to overlay machine data.ppt
- Many videos posted on <https://videoatwork.cat.com>
 - o https://videoatwork.cat.com/viewerportal/cat/home.vp?programId=esc_program%3A31713&contentAssociationId=association%3A143021&clipTimeIn=10094&clipDuration=238169
 - o https://videoatwork.cat.com/viewerportal/cat/home.vp?programId=esc_program%3A23984&contentAssociationId=association%3A90718&clipTimeIn=8333&clipDuration=484100
- Haul Template 1.rtt
- Model comparison Template 1.rtt
- RaceRender software can be purchased at <http://racerender.com/Products/index.html>

6.0 Related Best Practices

7.0 Acknowledgements

- This Best Practice was written by Francis Molloy
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